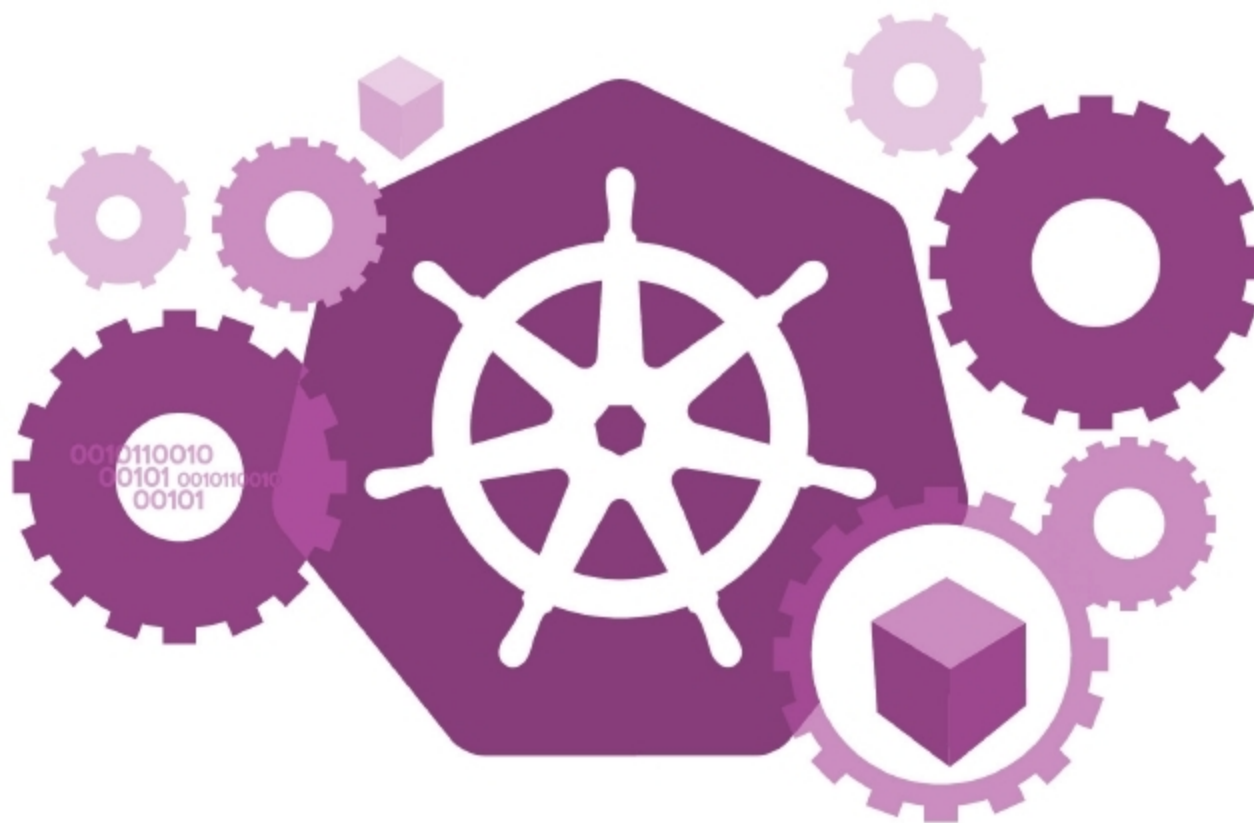


Demystifying Kubernetes

Kubernetes is a production grade open source system for automating deployment, scaling, and the management of containerised applications. This article is about managing containers with Kubernetes.



‘Containers’ has become one of the latest buzz words. But what does the term imply? Often associated with Docker, a container is defined as a standardised unit of software. Containers encapsulate the software and the environment required to run the software into a single unit that is easily shippable.

A container is a standard unit of software that packages the code and all its dependencies so that the application runs quickly and reliably from one computing environment to another. The container does this by creating something called an image, which is akin to an ISO image. A container image is a lightweight, standalone, executable package of software that includes everything needed to run an application -- code, runtime, system tools, system libraries and settings.

Container images become containers at runtime and, in the case

of Docker containers, images become containers when they run on a Docker engine. Containers isolate software from the environment and ensure that it works uniformly despite differences in instances across environments.

What is container management?

Container management is the process of organising, adding or replacing large numbers of software containers. Container management uses software to automate the process of creating, deploying and scaling containers. This gives rise to the need for container orchestration—a tool that automates the deployment, management, scaling, networking and availability of container based applications.

Kubernetes

Kubernetes is a portable, extensible, open source platform for managing containerised workloads and services,

and it facilitates both configuration and automation. It was originally developed by Google. It has a large, rapidly growing ecosystem. Kubernetes services, support, and tools are widely available.

Google open sourced the Kubernetes project in 2014. Kubernetes builds upon a decade and a half of experience that Google had with running production workloads at scale, combined with best-of-breed ideas and practices from the community, as well as the usage of declarative syntax.

Some of the common terminologies associated with the Kubernetes ecosystem are listed below.

Pods: A pod is the basic execution unit of a Kubernetes application – the smallest and simplest unit in the Kubernetes object model that you create or deploy. A pod represents processes running on a Kubernetes cluster.

A pod encapsulates the running container, storage, network IP